

CURRICULUM VITAE – KUZNETSOV ALEKSEY

PERSONAL INFORMATION

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SHORT SUMMARY

PhD in theoretical physics with 7+ years of experience in quantitative research, statistical modeling, distributed computing and multidimensional data analysis. Currently **Researcher at Huawei Russian Research Institute**, where I develop novel data-aware strategies for training and inference LLM acceleration, and **Researcher at Institute of Applied Physics**, where I develop quasilinear statistical models and simulations of plasma turbulence.

SKILLS

Programming languages: Python (PyTorch, Pandas, Megatron-LM), C++ (MPI, OMP, thread), MATLAB, R.

Mathematics and statistics: Probability Theory, Mathematical Statistics, Linear Algebra, Multi-dimensional Differential Calculus, Mathematical Analysis, Correlation Analysis, Spectral Analysis

Research: LLM Inference and Training Optimization Algorithms; Strategies of Data Compression with Negligible Quality Drop, Zero-Order Backpropagation Methods; Quasilinear Numerical Theory of Plasma Turbulence; Radio Wave Generation at the Exoplanet Bow Shock

Tools: LaTeX, Git, Docker, CI, Linux

EDUCATION

Lobachevsky University; faculty "Advanced School of General and Applied Physics"

Master's Degree in Physics, 2016-2022

IAP RAS; "Physics of Plasmas"

PhD in Physics. 2022-2026

WORK EXPERIENCE

Huawei, Lomonosov Research Institute, Nizhny Novgorod,
2025 - Present, Researcher

- Proposed and implemented compression strategies for acceleration LLM Training. Achieved up to 20% LLM end-to-end training acceleration without quality drop
- Identified statistically significant dependencies in multidimensional KV caches, weights, and activations through large-scale data analysis.
- Designed and implemented several original data transformation strategies based on own observations of the data. As a result, improved the compression up to the SOTA values (from 1.45x to 1.66x for BF16, from 1.37x to 1.48x for FP8, from 1.09x to 1.21x for HiF4 data)
- Proposed balanced strategy of reducing transmission overhead of sparsification indices during LLM Inference with quality drop near 0.8%

Institute of Applied Physics of the Russian Academy of Sciences, Nizhny Novgorod,
2018 - Present, Researcher

- Developed statistical models of plasma turbulence with using complex nonlinear multidimensional integro-differential equations in terms of the probability distributions.
- Designed a numerical finite-difference approximation for each model. Implemented multi-threaded algorithm (MPI, OMP, thread) in C++ to obtain numerical solutions.
- Developed and implemented algorithms for statistical, correlation, and spectral data analysis in MATLAB and Python to verify the models and conduct a physical analysis of the results.

ACHIEVEMENTS

Award for achievements in HiF4 weights compression and contribution to the start of the project based on invented compression strategies

Award for the personal contribution to KV-Cache compression project

Winner of the 2024 "PhD Student" individual grant from the "BASIS" Foundation for Theoretical Physics and Mathematics.

Winner of the 2024 academician G.A. Razuvaev scholarship competition for postgraduate students.

Winner of multiple grants from the Russian Science Foundation and the "BASIS" Foundation for Theoretical Physics and Mathematics as part of a research group.

Second place at the 2019 Open Volga Student Mathematics Olympiad and multiple prize-winning places at various school mathematics and physics olympiads.

KEY PUBLICATIONS AND CONFERENCES

Opentalks.AI. Belgrade, Serbia, February 19-20, 2026

ECLIM 2024. Lisbon, Portugal, September 16-20, 2024.

All-Russian Astronomical Conference. 2024. SAO RAS, Nizhny Arkhyz, Russia, August 25-31, 2024.

COSPAR, 44th Scientific Assembly. Athens, Greece, July 16-24, 2022.

IAUGA 2022. Busan, Republic of Korea, August 2-11, 2022.

Saturating Magnetic Field of Weibel Instability in Plasmas with Bi-Maxwellian and Bikappa Particle Distributions // Plasma Physics Reports, vol. 48, №9, P. 973-982

Quasilinear Simulation of the Development of Weibel Turbulence in Anisotropic Collisionless Plasma// JETP, 2024, vol. 137, №6, P. 966-985

On the analytical description of the nonlinear stage of the Weibel instability in collisionless anisotropic plasma// J. Plasma Phys., 2023, vol. 89, №6, P. 175890601

Acceleration of Electrons in Plasmospheres of Hot Jupiter-Type Exoplanets with a Relatively Weak Magnetic Field // Geomagnetism and Aeronomy, vol. 64, №7, P. 1014-1020